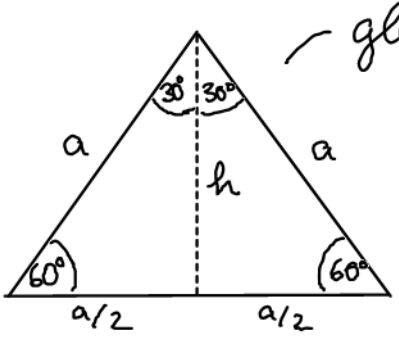
1.14 - Trigonometrische Funktionen



← gleichschenkliges Dreieck

$$\cos(60°) = \frac{a/2}{a} = 1/2$$

↙ Pythagoras

$$\sin(60°) = \frac{h}{a} = \frac{\sqrt{a^2 - (a/2)^2}}{a} = \frac{\sqrt{a^2(1-1/4)}}{a}$$

$$= \frac{a}{a} \cdot \sqrt{1-1/4} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

check: mit $\sin^2(\alpha) + \cos^2(\alpha) = 1$

$$\sin^2(60°) + \cos^2(60°) = \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{3}{4} + \frac{1}{4} = 1$$ ✓